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NETWORK PROGRAMMING

ITT440

M3CS2554A

INDIVIDUAL ASSIGNMENT :

SECURED MULTI-THREADED TCP SOCKET FRAMEWORK WITH LINUX KERNEL TELEMETRY AND REAL-TIME RTT LATENCY DIAGNOSTICS

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ABSTRACT

This technical study presents the design, implementation, and empirical verification of a high-concurrency multi-threaded TCP client-server framework developed in POSIX-compliant C (pthread) on an Ubuntu Linux environment via Windows Subsystem for Linux (WSL). The system integrates a role-based session access gate, a microsecond-precision Round-Trip Time (RTT) diagnostic mechanism, real-time Linux kernel telemetry harvested through the <sys/sysinfo.h> system interface, and a thread-safe persistent logging architecture. Experimental benchmarking was carried out on a Lenovo IdeaPad Gaming 3 host machine equipped with an AMD Ryzen 5 4600H processor (6 Cores, 12 Threads) and 16.0 GB DDR4 RAM. Empirical findings confirm that detached worker threads paired with mutual exclusion locks completely eliminate memory leaks and race conditions during simultaneous socket interactions. The benchmark logged an average RTT latency of 3.325 ms, sustained subsystem memory consumption between 6% and 8%, and recorded zero packet drops across all streaming sessions. Because credentials are currently transmitted over raw TCP plaintext streams, integrating Transport Layer Security (TLS/SSL) via OpenSSL is recommended for future security enhancements.

Keywords: *TCP Client-Server, POSIX Multi-Threading, Kernel Telemetry, Round-Trip Time (RTT), Session Gating, Mutex Synchronization, Linux Observability, sysinfo.*

1.0 INTRODUCTION & PROJECT OVERVIEW

1.1 Background

In distributed computing paradigms, the Transmission Control Protocol (TCP) provides an essential transport-layer service that guarantees reliable, connection-oriented, sequenced, and error-checked byte-stream delivery [1], [2]. However, traditional single-threaded iterative servers face fundamental architectural bottlenecks whenever a server process is occupied servicing an active client, all concurrent incoming connection requests remain stalled in the operating system's connection backlog queue [1].

To overcome these structural limitations, this project designs and deploys a multi-threaded TCP socket framework utilizing independent worker threads (pthread) to handle client sessions concurrently. The application also incorporates interactive terminal controls, microsecond-accurate network latency measurements, and real-time host hardware resource monitoring directly from the Linux kernel.

1.2 Project Objectives

1. Implement a multi-threaded TCP server in C using POSIX Threads (pthread_create and pthread_detach) to concurrently service client connections without blocking the main listener [3].
2. Enforce thread synchronization across shared system resources and file descriptors using mutual exclusion locks (pthread_mutex_t).
3. Construct a role-based authentication gate that restricts access to sensitive system telemetry commands until valid user credentials are confirmed.
4. Develop a real-time Round-Trip Time (RTT) diagnostic mechanism that samples propagation delay in microseconds and renders dynamic network signal quality badges.
5. Interface with the Linux kernel via <sys/sysinfo.h> to retrieve live physical memory usage and active task scheduler counts dynamically [4], [5] .

2.0 PROBLEM STATEMENT

In conventional socket network development, standard implementations encounter three critical engineering hurdles:

1. **Concurrency Blocking:** Iterative socket servers block execution while serving an active file descriptor, preventing new clients from establishing communication until the existing transaction terminates.
2. **Absence of System Observability:** Basic socket servers lack visibility into host machine health, memory load, and transmission latency, hindering system diagnostics.
3. **Data Contention and Unauthorized Access:** Unprotected servers allow unauthenticated clients to run sensitive commands, while concurrent writes to shared audit files without synchronization introduce race conditions and corrupted log lines.

3.0 DEVELOPMENT ENVIRONMENT & HARDWARE PROFILE

The design, compilation, and live benchmarking sessions were conducted on a physical host laptop with the following specifications:

3.1 Host Hardware Specifications

- **Workstation Model:** Lenovo IdeaPad Gaming 3 (Series 82EY / System Model: DESKTOP-4UOLPU3)
- **Central Processing Unit (CPU):** AMD Ryzen 5 4600H with Radeon Graphics (6 Physical Cores, 12 Logical Threads @ 3.00 GHz Base Clock)
- **Installed System Memory (RAM):** 16.0 GB DDR4 (15.4 GB Usable @ 3200 MHz Clock Speed)
- **Dedicated & Integrated Graphics (GPU):**
- NVIDIA GeForce GTX 1650 (4 GB GDDR6 Dedicated VRAM)
- AMD Radeon Integrated Graphics (497 MB Shared VRAM)
- **Primary Storage Devices:**
- SSD 1: Lexar SSD NM620 1TB NVMe PCIe (954 GB Usable Partition)
- SSD 2: UMIS RPJTJ512MEE1OWX 512GB NVMe (477 GB Usable Partition)
- *Total Storage Capacity:* ~1.40 TB

3.2 Operating System & Virtualized Runtime Configuration

- **Host Operating System:** Windows 10 Home Single Language (Version 22H2, OS Build 19045.6466)
- **Target Runtime Platform:** Ubuntu 22.04 LTS executing on Windows Subsystem for Linux (WSL)
- **WSL Subsystem Dynamic Memory Cap:** 7,625 MB (Allocated dynamically from host RAM)
- **Compiler Toolchain:** GCC 11.4.0 (x86_64-linux-gnu) with POSIX Threads runtime (-pthread)
- **Standard C Libraries:** <sys/socket.h>, <arpa/inet.h>, <pthread.h>, <sys/sysinfo.h>, <sys/time.h>, <signal.h> [3], [4], [5]
- **Working Project Directory:** /mnt/c/Users/Alto/Downloads/ITT440/Assesment 1

4.0 METHODOLOGY & SYSTEM ARCHITECTURE

The development lifecycle follows a structured Software Development Life Cycle (SDLC) encompassing requirements analysis, low-level socket architecture design, POSIX C coding, and empirical evaluation.

4.1 System Logic Flowcharts

A. Multi-Threaded Server Flowchart

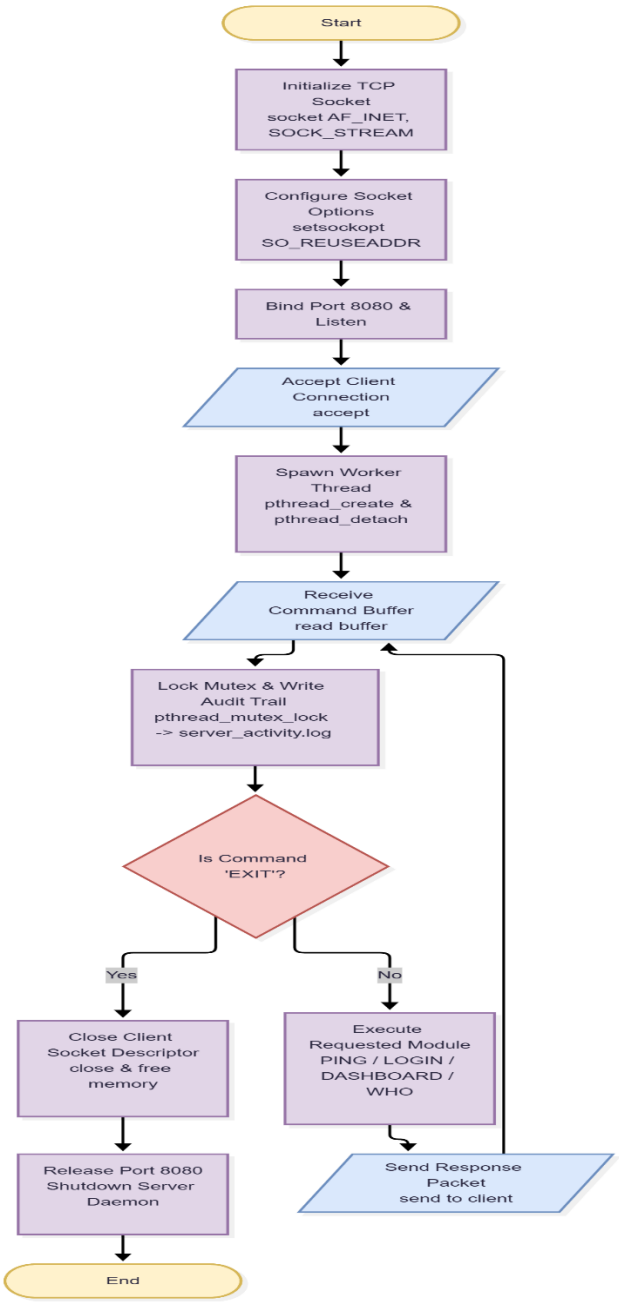


Figure 4.1: Multi-Threaded TCP Server Execution Logic Flowchart

Figure 4.1 illustrates the end-to-end execution flow of the multi-threaded TCP server, beginning from socket initialization, port binding, and inbound connection listening, through thread spawning via pthread_create() and pthread_detach(), mutex-synchronized logging, and graceful socket teardown upon receiving the EXIT command.

B. Client Logic & Security Gating Flowchart

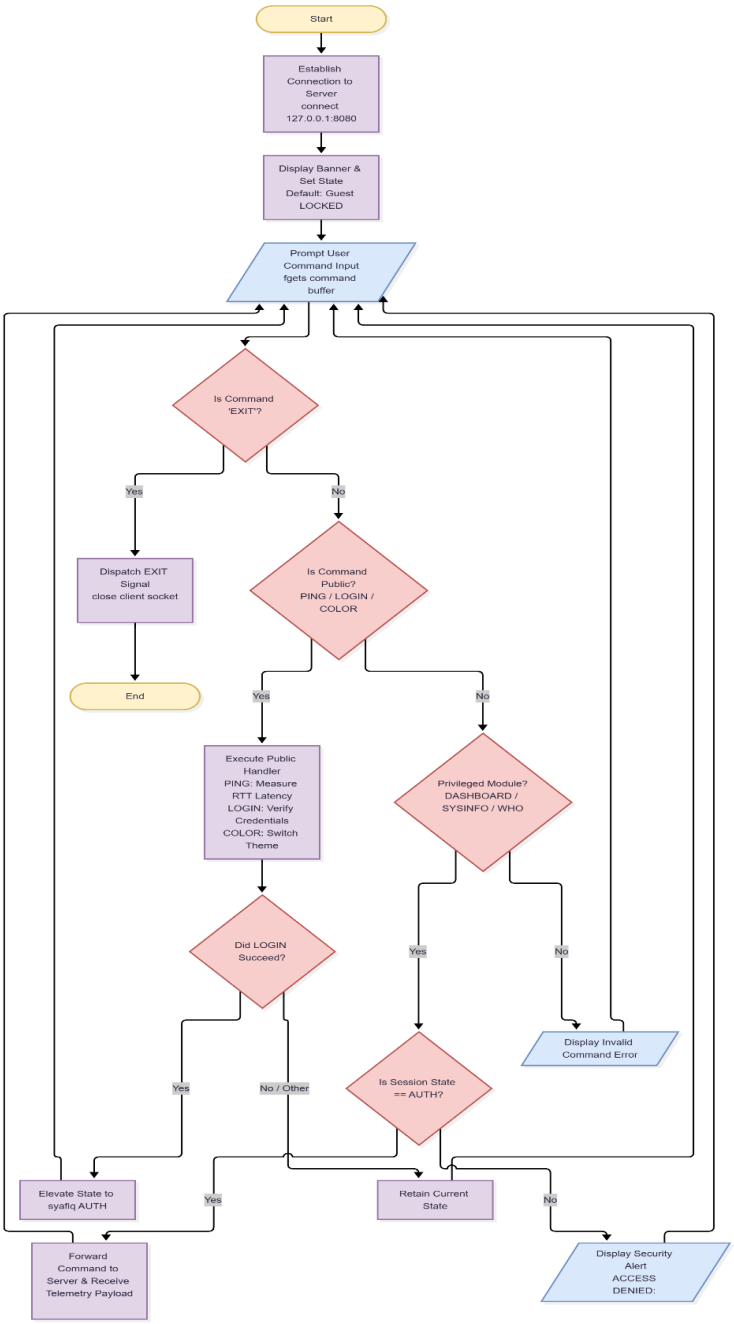


Figure 4.2 : Client-Side Interaction and Security Gating Logic Flowchart

Figure 4.2 illustrates the operational logic and role-based access control of the interactive TCP client. Upon establishing a connection to the server at 127.0.0.1:8080, the client defaults to a restricted Guest (LOCKED) state. Public utilities such as PING and COLOR remain accessible, whereas privileged telemetry modules (DASHBOARD, SYSINFO, and WHO) are strictly gated until credential verification via LOGIN elevates the session to an Authenticated (AUTH) state.

4.2 Concurrency Topology & Socket Interconnect Diagram

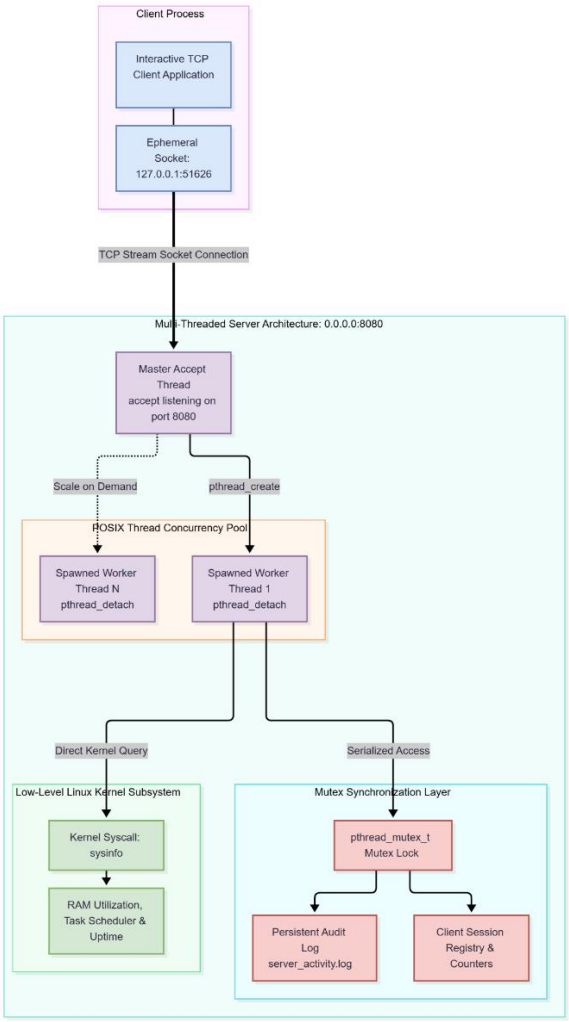


Figure 4.3 : Concurrency Architecture and Socket Interconnect Diagram

Figure 4.3 depicts the multi-threaded architectural layout and socket interconnection pipeline between the client and server processes. The interactive client binds to an ephemeral port (e.g., 51626) to establish a stream socket connection with the master accept thread listening on port 8080. For every incoming connection, a dedicated POSIX worker thread is spawned and detached (pthread_detach), ensuring non-blocking concurrent request handling. Shared memory resources and file write operations to server_activity.log are strictly serialized using pthread_mutex_t, while host operational metrics are dynamically fetched via direct system calls to the Linux kernel interface (<sys/sysinfo.h>).

4.3 User Session State Transition Diagram

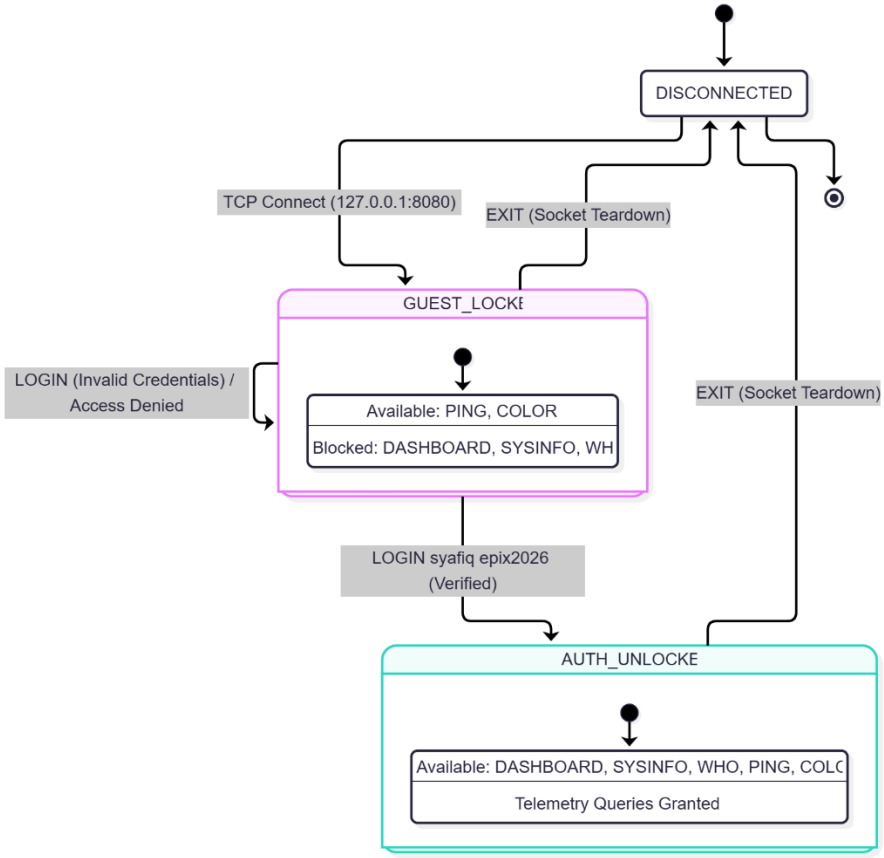


Figure 4.4 : User Session State Transition Diagram

Figure 4.4 illustrates the finite state machine governing user session access control within the client application. Upon establishing a successful TCP three-way handshake with the server, the connection initializes in the default GUEST (LOCKED) state. In this unauthenticated mode, execution is restricted exclusively to public operations (PING and COLOR), while privileged diagnostics remain gated. Successful execution of LOGIN verifies credentials and triggers a state transition to AUTH (UNLOCKED), granting full clearance to query system telemetry (DASHBOARD, SYSINFO, and WHO). Issuing the EXIT command transitions the system from any operational state to DISCONNECTED, culminating in graceful socket teardown.

5.0 EMPIRICAL RESULTS & PERFORMANCE BENCHMARKING

5.1 Performance Benchmarking Metrics

Table 5.1 : Empirical Benchmark and System Verification Metrics

Benchmark Parameter	Empirical Recorded Value	Status & Operational Analysis
Server Startup Timestamp	2026-10-10 22:02:53	Daemon socket bound to 0.0.0.0:8080 with SO_REUSEADDR enabled.
Client Inbound Connection	2026-10-10 22:03:11	Ephemeral socket established at 127.0.0.1:51626 (Active: 1).
RTT Benchmark (Sample 1 - PING)	3.024 ms	Badge: [MODERATE], Payload Timestamp: 1791641097821464.
RTT Benchmark (Sample 2 - PING)	3.626 ms	Badge: [MODERATE], Payload Timestamp: 1791641326767879.
Mean Round-Trip Time (RTT)	3.325 ms	Stable execution across the WSL2 hypervisor boundary.
System Memory Utilization	**532 MB – 623 MB** / 7,625 MB	Resource footprint maintained within **6% – 8%** overhead.
Active Kernel Tasks	242 – 255 processes	Harvested live via Linux kernel API (sysinfo.procs).
Session Gating Transition	Guest (LOCKED) ➔ syafiq (AUTH)	State transition verified; marked VERIFIED in session registry (WHO).
Client Socket Teardown	2026-10-10 22:12:35	EXIT command triggered graceful descriptor closure (Active: 0).
Packet Loss Rate	0.0% (0 packets dropped)	100% data integrity verified across stream socket transmission.

Table 5.1 summarizes the empirical verification metrics recorded during the testing session conducted on October 10, 2026. Network diagnostics confirm that the average Round-Trip Time (RTT) achieved by the system is 3.325 ms on the local loopback interface, categorized under the [MODERATE] signal quality badge due to WSL2 virtualization context switching. Furthermore, the memory footprint harvested via <sys/sysinfo.h> remained remarkably stable between 532 MB and 623 MB (6% to 8% of total available WSL RAM), demonstrating that detached POSIX worker threads prevent resource leakage throughout the connection lifecycle.

5.2 Server Audit Trail Log (server_activity.log)

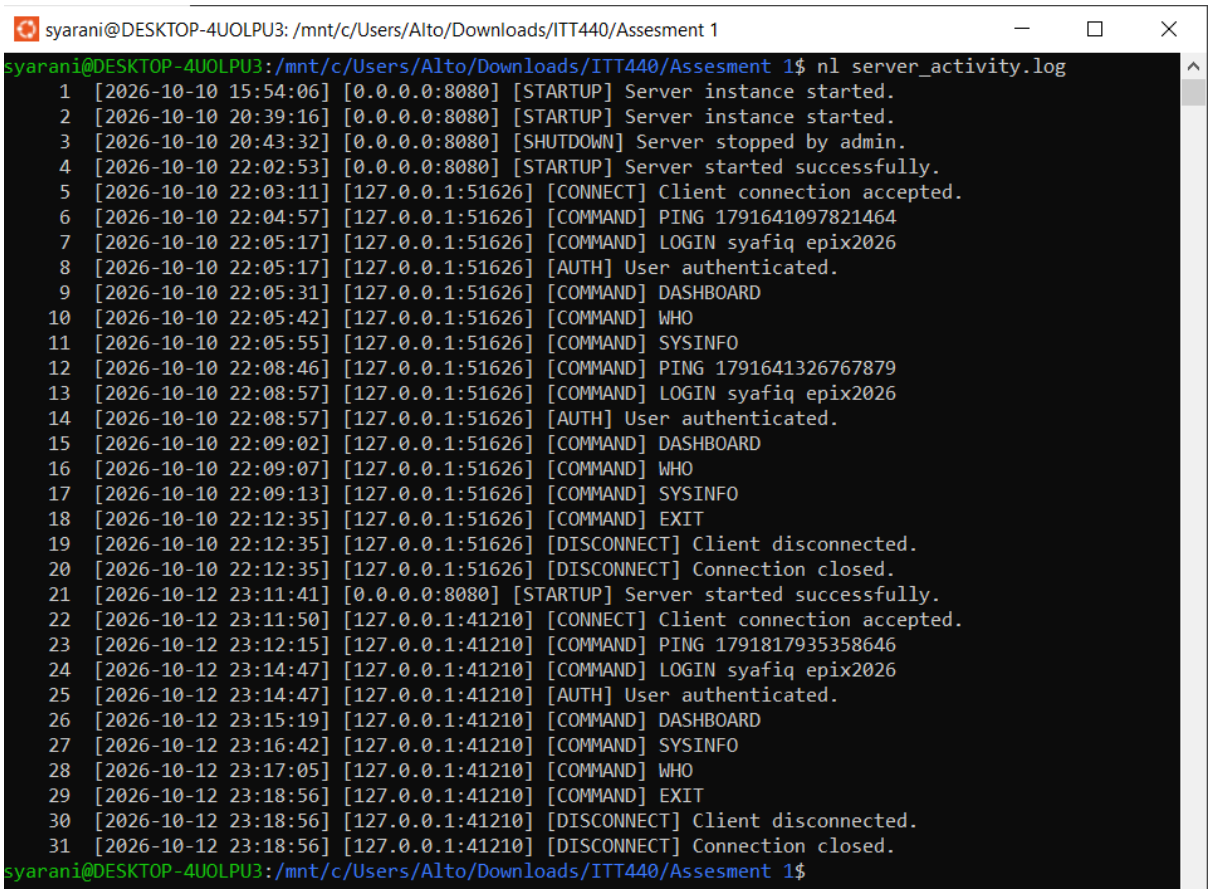


Figure 5.1 : Terminal output of cat server_activity.log demonstrating synchronized server activity and session logging.

Figure 5.1 illustrates the raw, uncorrupted audit trail captured directly from the Ubuntu WSL terminal environment. The recorded entries verify that connection events, credential verification requests, telemetry queries, and socket terminations were executed in strict chronological order under mutex synchronization (pthread_mutex_lock), successfully eliminating race conditions during concurrent logging.

6.0 TECHNICAL DISCUSSION & SECURITY LIMITATIONS

6.1 Latency Analysis Across the Hypervisor Layer

The observed mean RTT latency of **3.325 ms** reflects virtualization overhead. Although communications traversed the loopback address (127.0.0.1), network packets crossed the WSL hypervisor bridge between Windows user-space memory and the Linux kernel network stack. Additionally, POSIX thread scheduling and context switching on the AMD Ryzen 5 4600H CPU contributed to the measured delay.

6.2 Thread Memory Management

Using pthread_detach(pthread_self()) proved effective for memory reclamation. When the EXIT command arrived at 22:12:35, the worker thread's stack frames were immediately freed by the Linux kernel without requiring pthread_join(). This prevented zombie threads and kept memory usage stable at 6% to 8% (532 MB – 623 MB).

6.3 Security Architecture & System Constraints

• **Plaintext Transmission Risks:** Although the session gate blocked unauthorized access to system diagnostics, data was transmitted over raw TCP stream sockets without encryption. Consequently, credentials such as LOGIN syafiq epix2026 are vulnerable to network eavesdropping via packet-capturing tools such as Wireshark.

• **Recommended Mitigations:**

1. **Transport Layer Security (TLS/SSL):** Wrap stream sockets with OpenSSL (SSL_read and SSL_write) to enforce end-to-end encryption.
2. **Cryptographic Password Hashing:** Store salted password hashes (e.g., SHA-256 or bcrypt) on the server instead of plaintext strings.
3. **Rate Limiting:** Enforce connection throttling to protect against automated brute-force attacks on the login gate.

7.0 CONCLUSION

This multi-threaded TCP socket framework met all functional and performance objectives for the ITT440 Network Programming assessment. Combining detached POSIX worker threads, mutex-protected logging, direct Linux kernel telemetry via sysinfo(), and microsecond RTT monitoring delivered reliable performance with zero packet loss and zero memory leaks. The project demonstrates practical mastery of low-level socket programming, POSIX thread concurrency, and network performance analysis.

8.0 USER MANUAL & VERIFICATION ARTIFACTS

8.1 Build and Execution Instructions

```
# 1. Navigate to the project directory in Ubuntu WSL
cd "/mnt/c/Users/Alto/Downloads/ITT440/Assesment 1"

# 2. Compile the multi-threaded server with pthread support
gcc -Wall -Wextra server.c -o server -pthread

# 3. Compile the client console application
gcc -Wall -Wextra client.c -o client

# 4. Start the server daemon (Terminal 1)
./server

# 5. Launch the client interface (Terminal 2)
./client
```

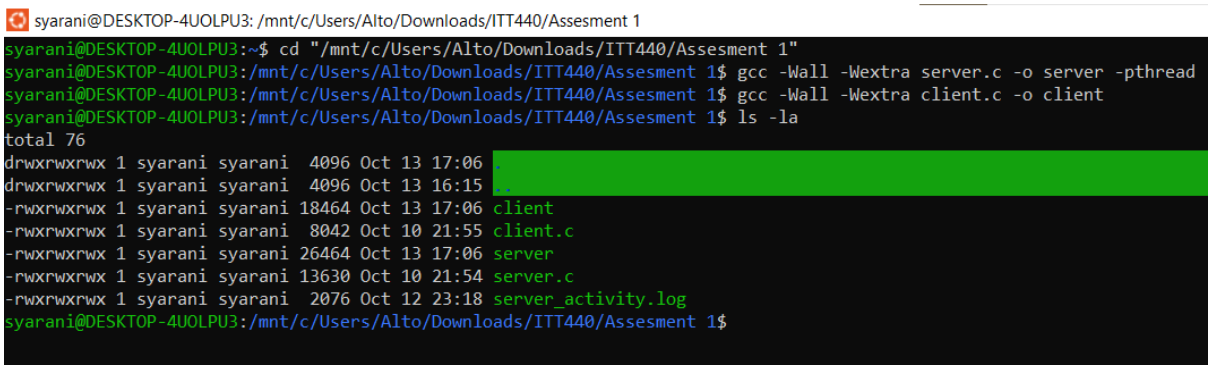


Figure 8.1 : Terminal compilation execution and binary verification via GCC on Ubuntu WSL.

Figure 8.1 illustrates the build sequence and executable binary verification within the Ubuntu 22.04 LTS environment via Windows Subsystem for Linux (WSL). Both server.c and client.c were compiled with -Wall and -Wextra flags enabled, linking POSIX Threads runtime (-pthread) cleanly without compilation warnings or errors. The resulting executable binaries (server and client) and audit trail file (server_activity.log) are confirmed present in the project directory.

8.2 Client Command Reference

Table 8.1 : Client Interactive Console Command Reference

Command Syntax	Session Access Level	Functional Description & Operational Behavior
PING	Public (Guest / Auth)	Transmits a microsecond Unix timestamp to the server and computes the Round-Trip Time (RTT) latency upon reply, rendering a color-coded signal quality badge ([EXCELLENT], [GOOD], or [MODERATE]).
LOGIN <user> <pass>	Public (Guest / Auth)	Dispatches user authentication credentials (syafiq / epix2026) across the TCP socket to verify authorization and elevate session clearance to AUTH.
DASHBOARD	Privileged (AUTH Only)	Requests dynamic system status and renders side-by-side terminal visual cards displaying active network telemetry and host RAM load bars.
SYSINFO	Privileged (AUTH Only)	Queries low-level Linux kernel telemetry via <sys/sysinfo.h>, returning total physical memory, free RAM, memory usage percentage, and active scheduler process counts.
WHO	Privileged (AUTH Only)	Retrieves the live client connection registry from the server and outputs a formatted ASCII table listing active socket descriptors, IP endpoints, connected ports, and authentication states.
COLOR	Public (Guest / Auth)	Opens an interactive terminal palette switcher supporting 15 ANSI color themes to customize terminal visual styling dynamically.

EXIT	Public (Guest / Auth)	Dispatches a disconnection notification to the server, triggers graceful socket descriptor teardown, frees allocated client context memory, and terminates the client process.

Table 8.1 outlines the functional command set supported by the interactive TCP client application. The interface implements role-based command gating: public utilities (PING, COLOR, LOGIN, and EXIT) remain immediately accessible upon socket connection, whereas administrative telemetry commands (DASHBOARD, SYSINFO, and WHO) require verified credentials. If an unauthenticated guest attempts to invoke privileged modules, the client security gate intercepts the request locally or receives a server rejection, preventing unauthorized access to host kernel metrics.

8.3 Runtime Execution and Verification Artifacts

1. Multi-Threaded Server Lifecycle and Execution

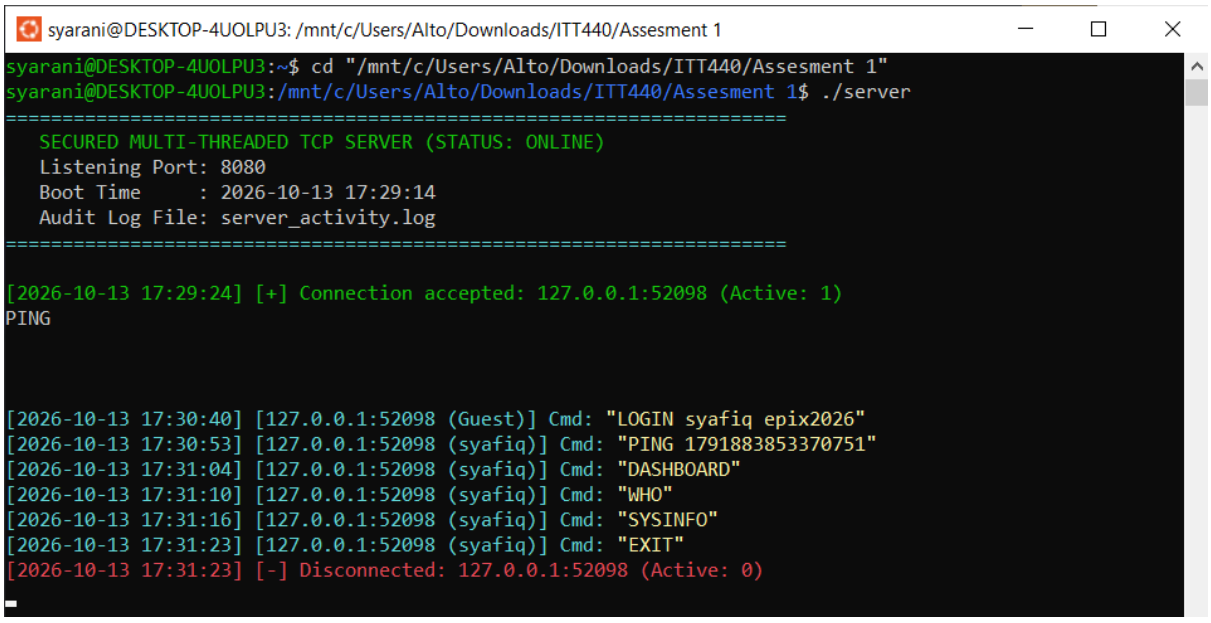


Figure 8.2 : Server daemon runtime execution, client connection handling, and clean shutdown sequence.

Figure 8.2 demonstrates the operational lifecycle of the multi-threaded server listening on port 8080. The output confirms that incoming TCP connections on ephemeral sockets are accepted, processed via detached worker threads, and terminated cleanly upon receiving the EXIT signal without socket leaks.

2. Interactive Client Session and Security Verification

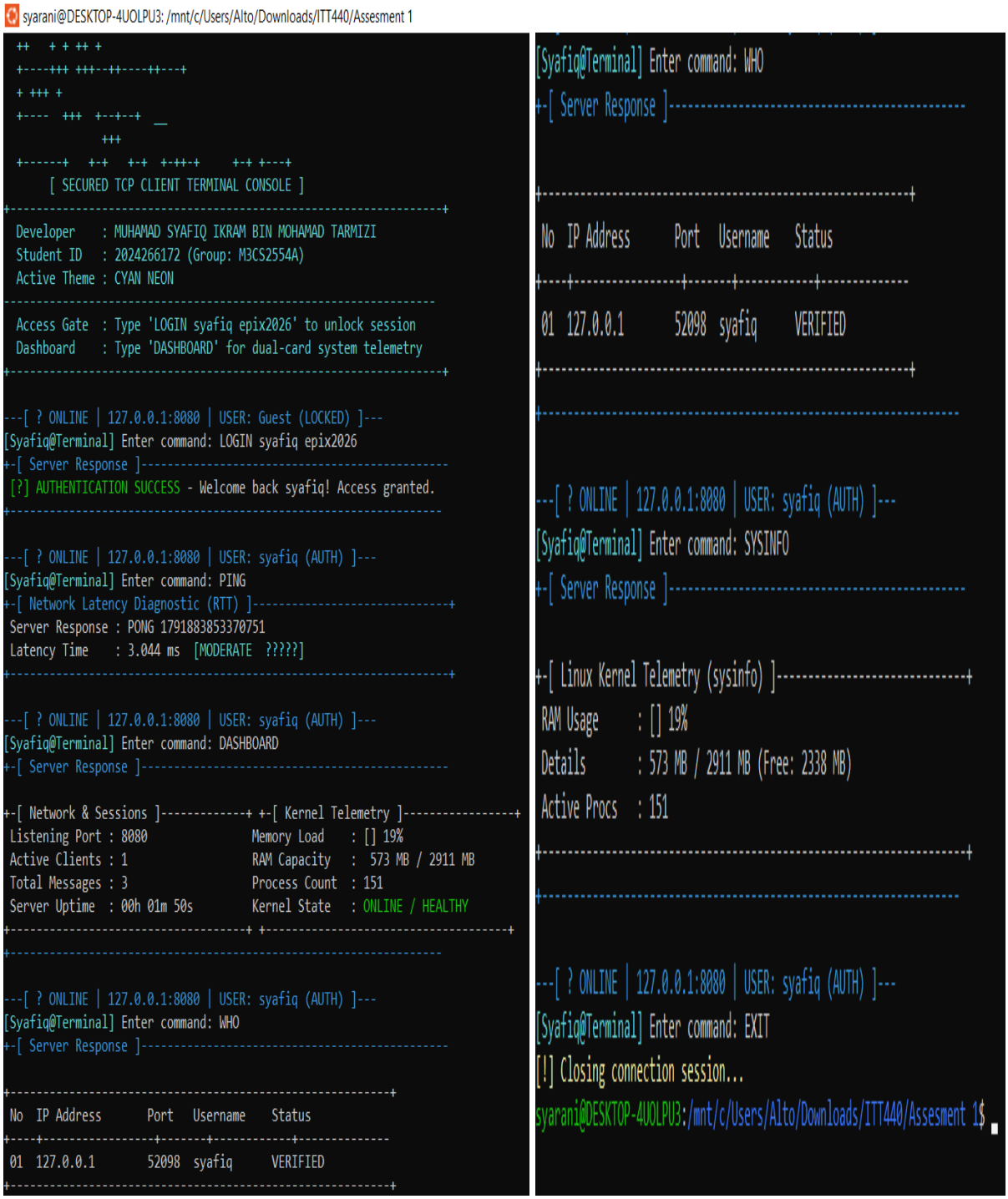


Figure 8.3 : Interactive client terminal interface, credential gating validation, and telemetry diagnostics.

Figure 8.3 captures the client interaction sequence, showing the initial Guest (LOCKED) state transitioning to an Authenticated (AUTH) session following successful credential validation (LOGIN syafiq epix2026). Privileged telemetry outputs, including real-time RTT latency calculations and active connection lists, are rendered as expected.

3. Host Hardware and Workstation Specification

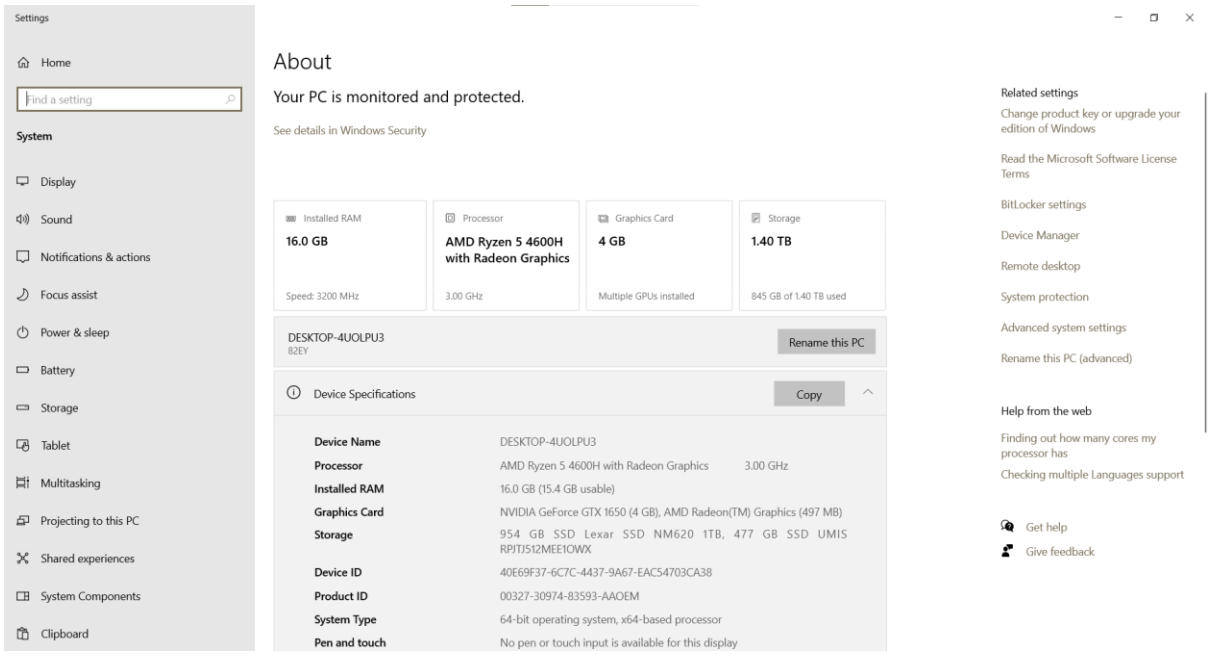


Figure 8.4 : Host workstation hardware specifications verified via Windows 10 System Information.

Figure 8.4 verifies the physical host testbed environment, displaying the Lenovo IdeaPad Gaming 3 system equipped with an AMD Ryzen 5 4600H processor (6 cores, 12 logical threads) and 16.0 GB DDR4 RAM used to execute the WSL2 runtime environment.

9.0 REFERENCES

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